

Multiple Choice

1. **Answer:** A

Options: A) 0.87×10^6 m/sec; B) 0.72×10^6 m/sec; C) 0.55×10^6 m/sec; D) 0.40×10^6 m/sec; E) 1.2×10^6 m/sec

Note: Correct option: A - 0.87×10^6 m/sec

2. **Answer:** C

Options: A) Dye laser; B) Semiconductor laser; C) Gas laser; D) Chemical laser; E) Fiber laser

Note: Correct option: C - Gas laser

3. **Answer:** A

Options: A) kinetic energy; B) number of electrons; C) potential energy; D) temperature of the nucleus; E) mass

Note: Correct option: A - kinetic energy

4. **Answer:** D

Options: A) It will quadruple.; B) It will increase by a factor of 2.; C) It will become zero.; D) It will be halved.; E) It will double.

Note: Correct option: D - It will be halved.

5. **Answer:** B

Options: A) 15 ohms; B) 1.5 ohms; C) 2.25 ohms; D) 50 ohms; E) 0.15 ohms

Note: Correct option: B - 1.5 ohms

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'quadruples'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'neutrons'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'All of these'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** magenta. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 6000 hp. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key